MINING DEALER BEST PRACTICE

Platforms for working on the TTT undercarriage

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

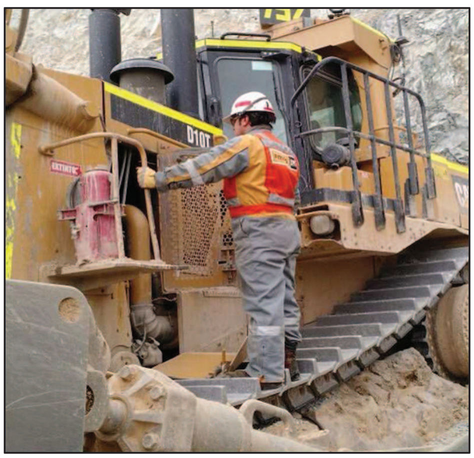
Auxiliary equipment such as the Track Type Tractors, have an important role in meeting the production targets of any mining operation. Maintenance and repair tasks must be completed in order to ensure the availability and reliability of the equipment, and these must be executed efficiently, with quality and safety.

There are a variety of maintenance and repair tasks associated with Track Type Tractors (TTT) that require the technicians to stand on the tracks. This Best Practice refers to the use of a specially designed platform(s) to be placed on the TTT's tracks, providing a safe work area for the technicians, which positively influences the quality and efficiency of the work.

2.0 Description of the Best Practice

2.1 Detection of the Opportunity

During the analysis of the technicians' exposure to risk while standing on the tracks of the TTT, three (3) large areas of opportunity for improvement were detected - safety, efficiency and quality of the work. The platform(s) to be installed on the track were designed to address these areas.



Areas of Opportunity

Safety

As shown in the photograph, the position that the technicians must adopt to work on the undercarriage, involves an inherent risk that we should be controlled and improved.

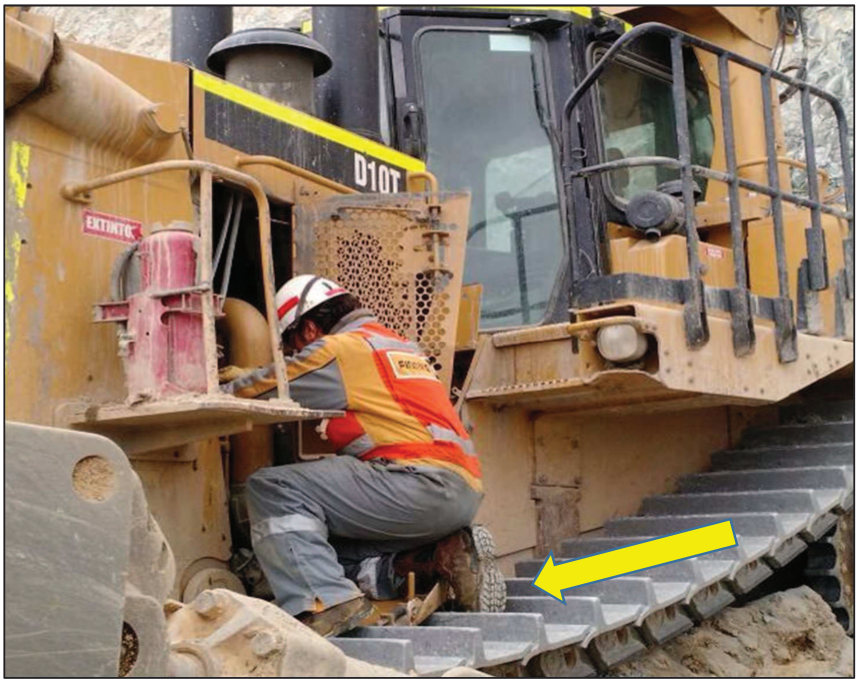
Efficiency or time to execute the work

Working in a risky area that is unstable and involves the possibility of slipping has a negative effect on the time the technicians will dedicate to execute the maintenance and repair tasks. Efficiency or time of execution has a direct influence on the machine's MTTR results and therefore on the fleet results of this type of equipment.

Quality or effectiveness of the work

The instability and risk involved with working on top of the tracks also affects the quality of the work done by the technicians. Our observations confirm this - technicians carried out sub-standard work when executing the tasks due to the lack of a solid and flat base on which to work.

The quality or effectiveness of the execution of the tasks has a direct influence on the machine's MTBF and MTBS.



Technician carrying out maintenance in areas above the undercarriage of a TTT (Track-Type Tractor)

2.2 Proposed Improvement Solution

The improvement solution proposed in this BP consists of the fabrication and use of specially designed platforms in order to provide the technician with a flat area where he can stand and move safely, when work must be done in the area above the TTT tracks.

Photograph of the work platform installed in a mining application D10T tractor (hard rock).

This type of platform provides our technicians with the necessary safety and stability to carry out work above the tracks with significantly increased control. This solution is a positive change to the original situation, which as we mentioned previously had inherent risks that also affected the quality and efficiency of the works.

Technicians executing Maintenance and Repair tasks on the tracks using the platforms proposed in this BP.

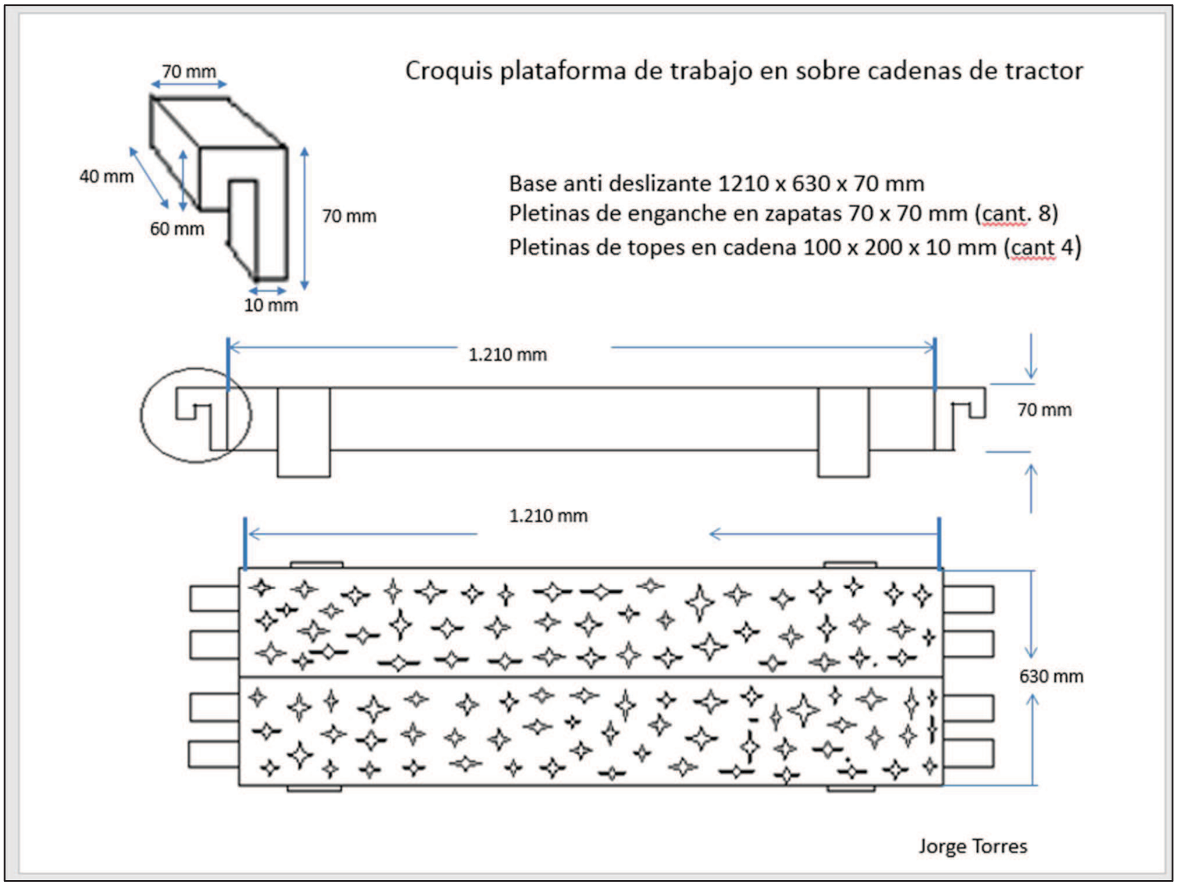
3.0 Steps for Implementation

The general steps that must be followed for the implementation of this Best Practice are:

3.1 Design and fabrication of the platform(s)
3.2 Practical tests and final adjustments
3.3 Documentation of the SOP (Standard Operating Procedure)
3.4 Personnel Training and change management.
3.5 Start-up and Validation of Benefits

3.1 Design and fabrication of the platform(s).

The design and fabrication of these platforms was done in our case by observing the specifications and dimensions of a D10T tractor, with application in hard rock mining. The same design concept must be used for the other TTT models, adapting the dimensions in accordance with the model and undercarriage of the machine. The appended diagram is for platforms to be used on D10T tractors.






Photographs of the work platforms from different angles.

3.2 Practical tests

Once the platforms have been fabricated, we recommend conducting tests on the equipment, preferably in the workshop, in a controlled and safe environment, with the objective of identifying and carrying out the final adjustments required.

3.3 Documentation of the SOP (Standard Operating Procedure)

Reviewing the safety regulations specific to usage of the platform is recommended in order to insure the platform follows all safety requirements. In order to ensure the proper usage and maintenance of this accessory, we recommend documenting the standard use procedure (SOP).

3.4 Employee Training and Change Management

It is important that employees who will use the work platforms routinely in maintenance and repair tasks on track-type tractors receive proper training.

We recommend that this training include the review of the standard procedure (SOP) as well as practice with its use in a safe and controlled environment.

The training and then the incorporation of the SOP into the databases and normal operation of the M&R routines play a key role in guiding employees to use the best practices (Change Management).

It is important to consider the construction of platforms for both workshop and field technicians. This is because there is a variety of M&R tasks that are carried out in the field and in the workshop on this type of equipment (TTT).

3.5 Start-up and Validation of Benefits

Once the platform has been put into operation, its correct use must be monitored and the benefits recorded.

Once the expected benefits have been obtained, the specific Standard Jobs must be improved / modified with the Planning and Scheduling Department, especially the new stopped machine time and man-hours.

4.0 Benefits

As we mentioned previously, the benefits associated with this BP are related to the areas of Safety, Efficiency and Quality.

Safety: The M&R technicians are provided with a solid base that is flat, stable and free from obstacles on which to carry out work when they have to position themselves over the tracks. The risk levels decrease significantly.

Efficiency & Quality: The time needed for the execution of the M&R work will be positively affected by using the proposed platforms. The technicians will work more efficiently and do better quality work when they do so in a safe and more comfortable position.

Efficiency or time of execution will have a direct influence on the machine's MTTR. It will also have a positive influence on the labor resources invested in the M&R tasks. The quality of the execution of the tasks has a direct influence on the machine's MTBF and MTBS results.

An improvement of the MTBS and MTTR results will have a positive effect on the availability results of the machine.

5.0 Resources Required

Those described in this BP associated with the materials and fabrication of the work platforms.

6.0 Additional Support / References

The fabrication diagram for the type of platform to be used for D10T tractors is appended to this BP. Remember that if this type of platform is to be applied to other tractor models or models equipped with another type of track, the dimensions proposed for the D10T tractors must be modified.
If additional information is required, please contact the author (s) of this BP directly.

7.0 Other Associated Best Practices

N/A

8.0 Acknowledgements

Generation, Implementation and Documentation of the Best Practice

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